

Packet Flow Analysis

1) TCP Three-Way Handshake

TCP is a connection-oriented protocol that ensures reliable communication between a client and a server. Before any data is exchanged, TCP establishes a connection using a three-step process called the **Three-Way Handshake**.

Steps of TCP Three-Way Handshake:

Step 1: SYN (Synchronize)

The client sends a SYN packet to the server, requesting to establish a connection.

Client-----SYN----->Server

Purpose:

- Initiates the connection
- Synchronizes sequence numbers

Step 2: SYN-ACK (Synchronize-Acknowledgement)

The server accepts the request and responds with SYN-ACK packet.

Client<-----SYN-ACK-----Server

Purpose:

- Acknowledges the client's request
- Sends the server's sequence number.

Step 3: ACK (Acknowledgement)

The client sends an ACK packet confirming receipt of the server's response.

Client-----ACK----->Server

Purpose:

- Completes the connection setup.
- Data transfer can now begin

Importance:

- Establishes a reliable connection
- Synchronizes sequence numbers
- Verifies that both devices are ready to communicate
- Prevents duplicate or incomplete connection

2) DNS Resolution Flow:

The Domain Name System (DNS) translates a human-readable domain name (e.g., www.google.com) into an IP address that computers use to communicate.

Without DNS, users would have to remember numerical IP addresses instead of website names.

DNS Resolution Steps:

Step 1: The user enters a website URL in the browser

Step 2: The browser checks its local cache to see if the IP address is already stored. If found, it immediately uses that IP address.

Step 3: If the IP is not available locally, the browser sends a DNS query to the configured DNS server.

Step 4: The DNS server looks up the IP address corresponding to the requested domain.

Step 5: The DNS server sends the IP address back to the client.

Step 6: The browser uses the returned IP address to establish a connection with the web server

Importance of DNS:

- Converts domain names into IP addresses
- Makes internet browsing user-friendly
- Reduces the need to memorize IP addresses
- Speeds up website access using DNS caching

3) HTTP Request-Response Flow

Hypertext Transfer Protocol (HTTP) is the application layer protocol used for communication between a web browser (client) and a web server.

The browser requests resources, and the server responds with the requested data.

HTTP Communication Steps:

Step 1: The user enters a website URL

Step 2: The browser sends an HTTP request

Step 3: The web server processes the request

Step 4: The server sends an HTTP response.

Step 5: The browser renders the webpage for the user

HTTPS Communication

Modern websites use HTTPS instead of HTTP.

HTTPS encrypts all communication using TLS (Transport Layer Security), ensuring that transmitted data remains secure and cannot be intercepted by unauthorized parties.